Yu-Jui Liang 梁栩睿

Algorithms / System Simulation / Testing & Validation

學歷：國立臺灣大學｜資料科學碩士學位學程（預計 2027 年 6 月畢業）
國立政治大學｜資訊科學系學士（雙主修：電子物理學士學位學程）
指導教授：江介宏教授（Jie-Hong Roland Jiang），國立臺灣大學
共同指導：鄭湘筠博士（Hsiang-Yun Cheng），中央研究院
專業領域：C++/Python, optimization, system simulation, GPU/memory systems, validation

01 / ALGORITHMS & OPTIMIZATION

Multi-Output Standard-Cell Technology Mapping with Differentiable Optimization

Purpose. Explore algorithmic improvements to a logic-synthesis mapping problem with correctness and measurable cost constraints.

Approach. Extended a C++ cut-based mapping codebase with multi-output discovery; integrated Boolean matching, library-based area/delay evaluation, and regression/correctness checks while preserving the original single-output matching path.

Evidence / Status. Evaluated 30 arithmetic benchmarks; reported a 2% average ADP improvement over the single-output baseline with mapping correctness maintained.

Relevant skills: C++, algorithm development, Boolean matching, regression validation, optimization, benchmarking

Discover Candidates ⟶ Evaluate Constraints ⟶ Validate / Benchmark

02 / GPU / MEMORY SYSTEMS (ONGOING)

Memory-Aware Hardware–Software Co-Design for Data Center LLM Serving

Purpose. Study efficient LLM serving in heterogeneous GPU systems, including prefill/decode disaggregation and hierarchical accelerator/host memory.

Approach. Investigating simulation-driven design-space exploration for resource scheduling, memory placement, data movement, and hardware–software configuration choices.

Evidence / Status. Ongoing research; performance evaluation and trade-off analysis remain investigation topics, with no quantified outcomes claimed here.

Relevant skills: GPU systems, memory architecture, resource scheduling, performance modeling, design-space exploration

System Configuration ⟶ Scheduling / Memory Policies ⟶ Trade-off Study

03 / LLM APPLICATIONS & FORMAL REASONING

SAT-LEX: LLM and SMT-Based Document Verification System

Purpose. Support more deterministic checking of domain-specific documents by combining extracted evidence with explicit rules and automated constraint solving.

Approach. Translated extracted document evidence and domain rules into structured constraints; applied an SMT solver to identify conflicting conditions and unsupported conclusions beyond unconstrained LLM responses.

Evidence / Status. The implemented constraint checks are intended to identify logical conflicts, inconsistent conditions, and unsupported conclusions.

Relevant skills: LLM applications, SMT / constraint solving, automated reasoning, verification workflows

Rules / Document Evidence ⟶ Structured Constraints ⟶ Solver Checks

04 / PROFESSIONAL SOFTWARE VALIDATION

System Simulation & Regression Validation – Neopower Technologies

Purpose. Improve test repeatability and problem diagnosis by reducing dependence on live-system testing alone.

Approach. Developed a simulator that reproduced operational and error scenarios; combined system-level testing, regression validation, post-deployment checks, customer-side A/B testing, and log/data analysis.

Evidence / Status. Professional engineering experience (Jun. 2023 – Mar. 2025) involving repeatable testing, regression checks, and diagnosis of software behavior.

Relevant skills: System simulation, software testing, regression validation, debugging, A/B testing, log analysis

Scenario Simulation ⟶ Repeatable Tests ⟶ Debug / Release Validation